



UNIVERSITY OF SAN-JOSE RECOLETOS

Engineering Mathematical Society

A.Y. 2026 - 2027



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Inverse Trigonometric Functions

Practice Problems

1. $y = \frac{2}{\sqrt{3}} \operatorname{arccot} \frac{2x+3}{\sqrt{3}}$

2. $y = \arctan \sqrt{e^x}$ at $x = 0$

3. $y = \arcsin \frac{2z}{1+x^2}$ at $x = 2$

4. $y = \frac{1}{\sin^{-1}(x)}$

5. $y = x \tan^{-1} \sqrt{x}$

6. $y = 4 \tan^{-1} 3x^4$

7. $y = \sin(x) + x^2 \tan^{-1}(x)$

8. $y = 3 \cos^{-1}(x^2 + 0.5)$

9. $y = 2 \cos(x) + 6 \cos^{-1}(x)$

10. $y = 5x^6 - \sec^{-1}(x)$



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Inverse Trigonometric Functions

Step-by-step Solution

1. $y = \frac{2}{\sqrt{3}} \operatorname{arccot} \frac{2x+3}{\sqrt{3}}$

- Rule: Use $\frac{d}{dx} \operatorname{arccot}(u) = -\frac{1}{1+u^2} \cdot u'$

- Step 1: Let the inner function be:

$$u = \frac{2x+3}{\sqrt{3}}$$

$$u' = \frac{2}{\sqrt{3}}$$

- Step 2: Apply the chain rule formula:

$$y' = \frac{2}{\sqrt{3}} \left(-\frac{1}{1+\left(\frac{2x+3}{\sqrt{3}}\right)^2} \right) \left(\frac{2}{\sqrt{3}} \right)$$

$$y' = \frac{4}{3} \left(-\frac{1}{1+\left(\frac{2x+3}{\sqrt{3}}\right)^2} \right)$$

- Step 3: Square the u term in the denominator:

$$1 + \frac{(2x+3)^2}{3} = 1 + \frac{4x^2+12x+9}{3}$$

The equation becomes:

$$y' = -\frac{4}{3\left(1+\frac{4x^2+12x+9}{3}\right)}$$

- Step 4: Distribute the 3 into the denominator:

$$y' = -\frac{4}{3+4x^2+12x+9}$$

- Step 5: Simplify and factor:



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$$y' = -\frac{4}{3+4x^2+12x+9}$$

$$y' = -\frac{4}{4x^2+12x+12}$$

$$y' = -\frac{4}{4(x^2+3x+3)}$$

$$y' = -\frac{1}{(x^2+3x+3)}$$

Answer: $y' = -\frac{1}{(x^2+3x+3)}$

2. $y = \arctan\sqrt{e^x}$ at $x = 0$

- Rule: Use the derivative of $\arctan(u) = \frac{1}{1+u^2} \cdot u'$
- Step 1: Rewrite the inner function as:

$$\sqrt{e^x} = (e^x)^{\frac{1}{2}}$$

$$(e^x)^{\frac{1}{2}} = e^{\frac{x}{2}}$$

Then:

$$u = e^{\frac{x}{2}}$$

$$u' = \frac{1}{2}e^{\frac{x}{2}}$$

- Step 2: Apply the chain rule:

$$y' = \frac{1}{1+(e^{\frac{x}{2}})^2} \cdot \left(\frac{1}{2}e^{\frac{x}{2}}\right)$$

- Step 3: Simplify the formula:



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$$y' = \frac{1}{1+(e^{\frac{x}{2}})^2} \cdot (\frac{1}{2}e^{\frac{x}{2}})$$

$$y' = \frac{1}{1+e^x} \cdot (\frac{1}{2}e^{\frac{x}{2}})$$

- Step 4: Evaluate at $x = 0$:

$$y' = \frac{1}{1+e^x} \cdot (\frac{1}{2}e^{\frac{x}{2}})$$

$$f'(0) = \frac{1}{1+e^0} \cdot (\frac{1}{2}e^{\frac{0}{2}})$$

$$f'(0) = \frac{1}{1+1} \cdot \frac{1}{2} (1)$$

$$y' = \frac{1}{4}$$

Answer: $y' = \frac{1}{4}$

3. $y = \arcsin \frac{2x}{1+x^2}$ at $x = 2$

- Rule: Use the derivative of $\arcsin(u) = \frac{1}{\sqrt{1-u^2}} \cdot u'$

- Step 1: Use the quotient rule to find u' where:

$$u = \frac{2x}{1+x^2}$$

$$w = 2x$$

$$dw = 2$$

$$v = 1 + x^2$$

$$dv = 2x$$

$$u' = \frac{(1+x^2)(2)-(2x)(2x)}{(1+x^2)^2}$$



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$$u' = \frac{(2+2x^2)-(4x^2)}{(1+x^2)^2}$$

$$u' = \frac{(2-2x^2)}{(1+x^2)^2}$$

$$u' = \frac{2(1-x^2)}{(1+x^2)^2}$$

- Step 2: Find the denominator of the arcsin derivative:

$$\sqrt{1 - \left(\frac{2x}{1+x^2}\right)^2}$$

- Step 3: Find the common denominator to simplify the radical:

$$\sqrt{1 - \left(\frac{2x}{1+x^2}\right)^2} = \sqrt{1 - \frac{4x^2}{(1+x^2)^2}}$$

$$\sqrt{1 - \frac{4x^2}{(1+x^2)^2}} = \sqrt{\frac{(1+x^2)^2}{(1+x^2)^2} - \frac{4x^2}{(1+x^2)^2}}$$

$$\sqrt{\frac{(1+x^2)^2}{(1+x^2)^2} - \frac{4x^2}{(1+x^2)^2}} = \sqrt{\frac{(1+x^2)^2 - 4x^2}{(1+x^2)^2}}$$

$$\sqrt{\frac{(1+x^2)^2 - 4x^2}{(1+x^2)^2}} = \sqrt{\frac{1+2x^2+x^4-4x^2}{(1+x^2)^2}}$$

$$\sqrt{\frac{1+2x^2+x^4-4x^2}{(1+x^2)^2}} = \sqrt{\frac{x^4-2x^2+1}{(1+x^2)^2}}$$

$$\sqrt{\frac{x^4-2x^2+1}{(1+x^2)^2}} = \sqrt{\frac{(x^2-1)^2}{(1+x^2)^2}}$$

$$\sqrt{\frac{(x^2-1)^2}{(1+x^2)^2}} = \frac{(x^2-1)}{1+x^2}$$



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- Step 4: Combine to get the equation:

$$y' = \frac{1}{\frac{(x^2-1)}{1+x^2}} \left(\frac{2(1-x^2)}{(1+x^2)^2} \right)$$

$$y' = \frac{2(1-x^2)}{(x^2-1)(1+x^2)}$$

- Step 5: Evaluate at $x = 2$:

$$f'(x) = \frac{2(1-x^2)}{(x^2-1)(1+x^2)}$$

$$f'(x) = \frac{-2(x^2-1)}{(x^2-1)(1+x^2)}$$

$$f'(x) = \frac{-2}{(1+x^2)}$$

$$f'(2) = \frac{-2}{(1+2^2)}$$

$$f'(2) = \frac{-2}{5}$$

Answer: $y' = -\frac{2}{5}$

4. $y = \frac{1}{\sin^{-1}(x)}$

- Rule: Use the chain rule by rewriting the expression as negative exponent:

$$y = [\arcsin(x)]^{-1}$$

- Step 1: Apply the power rule to the outside:

$$y = - [\arcsin(x)]^{-2}$$

- Step 2: Multiply by the derivative of the inside function:



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$$(\arcsin(x)) = \frac{1}{\sqrt{1-x^2}}$$

- Step 3: Combine and rewrite with positive exponents:

$$y' = - [\arcsin(x)]^{-2} \cdot \frac{1}{\sqrt{1-x^2}}$$

$$y' = - \frac{[\arcsin(x)]^{-2}}{\sqrt{1-x^2}}$$

$$y' = - \frac{1}{[\arcsin(x)]^2 \sqrt{1-x^2}}$$

$$\text{Answer: } y' = - \frac{1}{[\arcsin(x)]^2 \sqrt{1-x^2}}$$

5. $y = x \tan^{-1} \sqrt{x}$

- Rule: Use the product rule.

$$\frac{d}{dx} [uv] = uv' + vu'$$

$$u = x$$

$$du = 1$$

$$v = \arctan(\sqrt{x})$$

$$dv = \frac{1}{1+(\sqrt{x})^2} \cdot \frac{1}{2\sqrt{x}} = \frac{1}{1+x} \cdot \frac{1}{2\sqrt{x}}$$

- Step 1: Apply the product rule formula:

$$y' = (x) \left(\frac{1}{(2\sqrt{x})(1+x)} \right) + (\arctan(\sqrt{x}))(1)$$

- Step 2: Simplify and rewrite:

$$y' = \left(\frac{x}{(2\sqrt{x})(1+x)} \right) + (\arctan(\sqrt{x}))$$

$$y' = \frac{x}{(2)(\sqrt{x})(1+x)} + \arctan(\sqrt{x})$$



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$$y' = \frac{\sqrt{x}}{(2)(1+x)} + \arctan(\sqrt{x})$$

$$y' = \frac{\sqrt{x}}{2+2x} + \arctan(\sqrt{x})$$

Answer: $y' = \frac{\sqrt{x}}{2+2x} + \arctan(\sqrt{x})$

6. $y = 4\tan^{-1}3x^4$

- Rule: Use the chain rule with constant multiple.
- Step 1: Take the derivative of the outer function:

Outer function: $(\tan^{-1}(3x^4))$

$$y = (4)(\tan^{-1}(3x^4))$$

$$y' = (4)\left(\frac{1}{1+(3x^4)^2}\right) \cdot u'$$

$$y' = \frac{4}{1+(3x^4)^2} \cdot u'$$

- Step 2: Take the derivative of the inner function and combine:

$$u = 3x^4$$

$$u' = 12x^3$$

$$y' = \frac{4}{1+(3x^4)^2} \cdot 12x^3$$

$$y' = \frac{48x^3}{1+(3x^4)^2}$$

Answer: $y' = \frac{48x^3}{1+(3x^4)^2}$



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7. $y = \sin(x) + x^2 \tan^{-1}(x)$

- Rule: Use the standard differentiation for $\sin(u)$ and $\arctan(u)$.
- Step 1: Differentiate the first term:

$$\frac{d}{dx} [\sin(x)] = \cos(x)$$

- Step 2: Differentiate the second term using the product rule:

$$\frac{d}{dx} [uv] = uv' + vu'$$

$$u = x^2$$

$$du = 2x$$

$$v = \arctan(x)$$

$$dv = \frac{1}{1+(x)^2} = \frac{1}{1+x^2}$$

$$\frac{d}{dx} [uv] = (x^2)\left(\frac{1}{1+x^2}\right) + (\arctan(x))(2x)$$

Simplify:

$$\frac{d}{dx} [uv] = \left(\frac{x^2}{1+x^2}\right) + 2x \arctan(x)$$

- Step 3: Combine the derivatives:

$$y' = \cos(x) + \left(\frac{x^2}{1+x^2}\right) + 2x \arctan(x)$$

Answer: $y' = \cos(x) + \left(\frac{x^2}{1+x^2}\right) + 2x \arctan(x)$



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8. $y = 3\cos^{-1}(x^2 + 0.5)$

- Rule: Use the standard derivative of $\cos^{-1}(u)$:

$$\cos^{-1}(u) = -\frac{1}{\sqrt{1-u^2}} \cdot u'$$

- Step 1: Take the derivative of the outer function:

$$y' = 3\left(-\frac{1}{\sqrt{1-u^2}} \cdot u'\right)$$

- Step 2: Take the derivative of the inner function:

$$u = x^2 + 0.5$$

$$u' = 2x$$

- Step 3: Rewrite and simplify the equation:

$$y' = 3\left(-\frac{1}{\sqrt{1-u^2}} \cdot u'\right)$$

$$y' = 3\left(-\frac{1}{\sqrt{1-(x^2+0.5)^2}} \cdot 2x\right)$$

$$y' = -\frac{6x}{\sqrt{1-(x^4+x^2+0.25)}}$$

$$\text{Answer: } y' = -\frac{6x}{\sqrt{-x^4-x^2+0.75}}$$



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9. $y = 2\cos(x) + 6\cos^{-1}(x)$

- Rule: Differentiate each term individually.
- Step 1: Take the derivative of the first term:

$$\frac{d}{dx} [2\cos(x)] = -2\sin(x)$$

- Step 2: Take the derivative of the second term:

$$\frac{d}{dx} [6\cos^{-1}(x)] = 6\left(-\frac{1}{\sqrt{1-x^2}}\right)$$

- Step 3: Combine to get the final answer:

$$y' = -2\sin(x) - \frac{6}{\sqrt{1-x^2}}$$

Answer: $y' = -2\sin(x) - \frac{6}{\sqrt{1-x^2}}$

10. $y = 5x^6 - \sec^{-1}(x)$

- Rule: Apply the power rule to the first term and the standard derivative rule for the inverse secant.

- Step 1: Take the derivative of the first term:

$$\frac{d}{dx} [5x^6] = 30x^5$$

- Step 2: Take the derivative of the second term:

$$\frac{d}{dx} [\sec^{-1}(x)] = \frac{1}{|x|\sqrt{x^2-1}}$$

- Step 3: Combine the terms:



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$$y' = 30x^5 - \frac{1}{|x|\sqrt{x^2-1}}$$

Answer: $y' = 30x^5 - \frac{1}{|x|\sqrt{x^2-1}}$